# DGKE-Associated Atypical Hemolytic Uremic Syndrome — Comprehensive Disease Report

**MONDO:** MONDO:0100590 · **OMIM:** #615008 (nephrotic syndrome type 7 / aHUS 7) · **Causal gene: DGKE** (diacylglycerol kinase epsilon; HGNC:2852) · **Category:** Mendelian (autosomal recessive)

## Summary

**DGKE-associated atypical hemolytic uremic syndrome (aHUS) is an autosomal-recessive, complement-independent thrombotic microangiopathy caused by biallelic loss-of-function variants in *DGKE*, the gene encoding diacylglycerol kinase epsilon (DGKε).** This distinguishes it mechanistically from the majority of aHUS cases, which arise from dysregulation of the alternative complement pathway. The causal gene — to state it explicitly in line with the research brief — is **DGKE**, a lipid kinase with *no* role in the complement cascade. The seminal exome-sequencing study of Lemaire and colleagues identified recessive *DGKE* mutations co-segregating with aHUS in nine unrelated kindreds, defining a distinctive Mendelian disease (PMID: 23542698).

DGKε is a small, membrane-bound, constitutively active lipid kinase that is unique among the ten mammalian DGK isoforms in its specificity for 1-stearoyl-2-arachidonoyl-*sn*-glycerol — the diacylglycerol (DAG) backbone of the phosphatidylinositol (PI) cycle. Loss of enzyme function leaves arachidonoyl-DAG elevated, which activates protein kinase C (PKC) and p38-MAPK to generate a **prothrombotic, repair-impaired endothelial state**, and simultaneously starves VEGFR2 signaling of phosphatidylinositol-4,5-bisphosphate (PIP2), collapsing the downstream Akt→COX-2→prostaglandin E2 (PGE2) axis that normally protects the glomerular microvasculature. The net result is microvascular thrombosis and podocyte/endothelial injury in the kidney.

Clinically, the disease presents in infancy (typically before 1–2 years of age) with the classic TMA triad (microangiopathic hemolytic anemia, thrombocytopenia, acute kidney injury) *plus* a distinctive nephropathy — nephrotic-range proteinuria, hematuria, and hypertension — frequently with **normal complement C3**. The course is relapsing and progresses to chronic kidney disease (CKD)/end-stage kidney disease (ESKD) in a substantial fraction of patients. Because the disease is not complement-driven, anti-C5 therapy (eculizumab) is frequently ineffective at preventing relapses, and genetic diagnosis supports de-escalation of complement blockade in favor of supportive, nephroprotective care. Restoration of the PGE2 axis reversed disease in endothelial-specific *Dgke*-knockout mice and represents the most promising mechanism-based therapeutic lead.

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## Section 1 — Disease Information

**Overview.** DGKE-associated aHUS is an ultra-rare, infantile-onset thrombotic microangiopathy affecting principally the kidney. It belongs to the broader aHUS family but is pathogenically distinct: it is caused by loss of an intracellular lipid kinase rather than by complement dysregulation. The phenotypic spectrum overlaps with membranoproliferative glomerulonephritis (MPGN)-like injury and with isolated steroid-resistant nephrotic syndrome (SRNS), such that the same *DGKE* genotype can present as aHUS, MPGN, or SRNS (PMID: 32413569; PMID: 32386968).

**Key identifiers.**

| Resource | Identifier |
|---|---|
| MONDO | MONDO:0100590 |
| OMIM phenotype | #615008 (nephrotic syndrome, type 7; historically "aHUS susceptibility, type 7") |
| Causal gene | *DGKE* (HGNC:2852; NCBI Gene 8526; Ensembl ENSG00000153747) |
| Orphanet | aHUS / rare renal TMA (ORPHA:2134 family) |
| ICD-10 / ICD-11 | D59.3 (hemolytic-uremic syndrome) / 3A21.2 |
| MeSH | Atypical Hemolytic Uremic Syndrome |

**Synonyms / alternative names.** aHUS type 7; DGKE nephropathy; DGKE-associated thrombotic microangiopathy; diacylglycerol kinase epsilon nephropathy.

**Information source type.** Evidence is derived from *aggregated disease-level resources* — primarily small, international case series, national cohorts/registries, and mechanistic model-organism and in vitro studies — rather than from large individual-patient EHR datasets, reflecting the disease's rarity.

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## Section 2 — Etiology

**Primary cause (genetic).** Biallelic (homozygous or compound-heterozygous) loss-of-function variants in *DGKE*. Lemaire et al. identified recessive *DGKE* mutations in nine unrelated kindreds (PMID: 23542698). This is a monogenic, fully genetic etiology; there is no established infectious or primary environmental cause.

**Genetic risk factors / modifiers.** The disease itself is recessive and Mendelian, but *co-inherited complement variants* can modify onset and severity. In the Spanish aHUS registry, among patients with *DGKE* mutations, several also carried heterozygous variants in thrombomodulin (*THBD*) or *C3*, and complement dysregulation appeared to "influence the onset and disease severity in carriers of diacylglycerol kinase-ε mutations" (PMID: 25135762). Thus complement genes act as potential *modifiers* rather than primary causes.

**Environmental / triggering factors.** As in other TMAs, intercurrent infections and, in reported cases, pregnancy can act as triggers or relapse precipitants; one DGKE mutation was identified in a pregnancy-related TMA cohort (PMID: 40306205). Age (infancy) is the dominant demographic risk window.

**Protective factors.** No validated genetic or environmental protective factors have been defined for this ultra-rare monogenic disease. *Information not available.*

**Gene–environment interactions.** The principal documented interaction is between the causal *DGKE* genotype and *modifier complement genotype/triggers*, which together shape clinical expressivity (PMID: 25135762).

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## Section 3 — Phenotypes

Core phenotypes, with suggested HPO terms, drawn from Lemaire 2013 and the nationwide DGKε nephropathy cohort (n=12; PMID: 36988693):

| Phenotype | Type | HPO term | Frequency (cohort) | Onset / course |
|---|---|---|---|---|
| Microangiopathic hemolytic anemia | Lab abnormality | HP:0001937 (thrombotic microangiopathy); HP:0004809 | 11/12 | Infantile; episodic |
| Thrombocytopenia | Lab abnormality | HP:0001873 | common | Infantile; episodic |
| Acute kidney injury | Clinical sign | HP:0001919 | common | Infantile |
| Nephrotic-range proteinuria | Lab abnormality | HP:0012593 / HP:0000100 | 10/12 | Persistent; relapsing |
| Microscopic hematuria | Lab abnormality | HP:0000790 | 10/12 | Persistent |
| Hypertension | Clinical sign | HP:0000822 | ~40% at follow-up | Persistent |
| Hypoalbuminemia | Lab abnormality | HP:0003073 | 6/12 | — |
| Hypercholesterolemia | Lab abnormality | HP:0003124 | 6/12 | — |
| Hypocomplementemia (low C3) | Lab abnormality | HP:0005339 | 4/12 (minority) | Variable |
| Chronic kidney disease / ESKD | Clinical outcome | HP:0012622 / HP:0003774 | 60% death-or-CKD at 3.3 y | Progressive |

**Characteristics.** Age of onset is **neonatal-to-infantile**: "Patients presented at a median age of 11 (7.5, 12.3) months; all were younger than 2 years" (PMID: 36988693); Lemaire noted affected individuals "present with aHUS before age 1 year" (PMID: 23542698). Severity is moderate-to-severe and variable; progression is **episodic/relapsing on a progressive background** ("relapses (66.7%)... death or progression to CKD (60%)"). A notable feature is that complement C3 is **typically normal**, unlike complement-mediated aHUS, though a minority show low C3.

**Quality-of-life impact.** Disease-specific QoL instruments have not been applied in this ultra-rare population, but the burden is substantial: lifelong CKD/dialysis dependence, growth and developmental impact of infantile kidney disease, relapse surveillance, and transplant considerations. *Formal QoL metrics not available.*

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## Section 4 — Genetic / Molecular Information

**Causal gene.** *DGKE* (chromosome 17q22), encoding diacylglycerol kinase epsilon, a 64-kDa lipid kinase. **Inheritance is autosomal recessive**; disease requires two loss-of-function alleles.

**Pathogenic variant spectrum.** Variants span the gene and include missense, frameshift, nonsense, and splice-site classes. In a 12-patient cohort, 13 homozygous/compound-heterozygous variants were reported (7 pathogenic, 3 likely pathogenic by ACMG/AMP criteria), 11 of which were novel (PMID: 36988693). Recurrent/illustrative alleles include:

- **c.966G>A, p.Trp322Ter** (nonsense) — recurrent; reported homozygous in SRNS and compound-heterozygous in aHUS (PMID: 32413569).
- **c.171delG, p.Ser58Alafs*111** (frameshift) (PMID: 32413569).
- **p.His536Qfs*16** (homozygous frameshift); **p.Trp322*/p.Pro498Arg** and **p.Gln248His/p.Gly484Glyfs*10** (compound heterozygous) (PMID: 25135762).

**Variant classification & frequency.** Pathogenic alleles are individually ultra-rare in gnomAD, consistent with a recessive, severe pediatric phenotype. **Origin is germline**; no somatic contribution is described.

**Functional consequence.** Variants are **loss-of-function** (enzyme-dead). Lemaire et al. inferred that "loss of DGKE function results in a prothrombotic state" (PMID: 23542698). Mechanistically this is a true loss-of-function (not dominant-negative), consistent with recessive inheritance.

**Modifier genes.** Heterozygous complement variants (*THBD*, *C3*, *CFI*) can co-occur and modify severity/treatment response (PMID: 25135762; PMID: 38374836).

**Epigenetic information / chromosomal abnormalities.** No disease-specific methylation signature or large-scale cytogenetic abnormality has been described. *Not applicable / not available.*

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## Section 5 — Environmental Information

- **Environmental toxins/pollution/radiation/occupational exposure:** none established. *Not applicable.*
- **Lifestyle factors:** none established (disease onsets in infancy). *Not applicable.*
- **Infectious agents:** The disease is *not* caused by Shiga-toxin–producing *E. coli* (which causes typical HUS). Infections may act as non-specific relapse triggers, as in other TMAs, but no specific pathogen is causal. The defining distinction from typical HUS is the absence of a Shiga-toxin etiology.

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## Section 6 — Mechanism / Pathophysiology

### Ordered causal chain (initiating lesion → clinical manifestation)

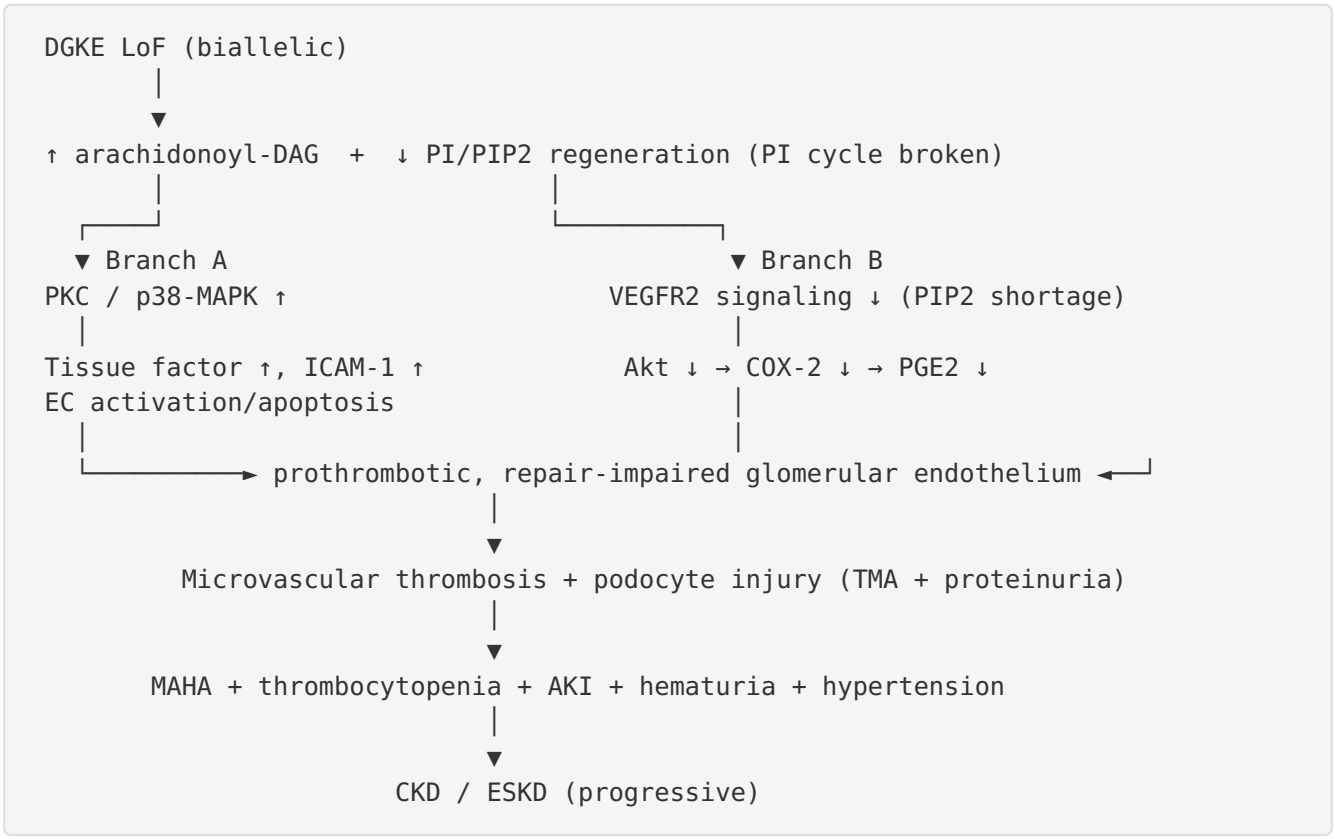
1. **Biallelic loss-of-function mutation in *DGKE*** *leads to* absent/non-functional diacylglycerol kinase epsilon protein in cells that normally express it (endothelium, podocytes, platelets). *(Demonstrated: PMID: 23542698.)*
2. Loss of the only arachidonoyl-DAG–specific kinase *results in* **accumulation of 1-stearoyl-2-arachidonoyl-DAG** and **failure to regenerate phosphatidylinositol/PIP2** via the PI cycle. *(Demonstrated biochemically: PMID: 27803897; PMID: 38008259.)*

**Branch A — prothrombotic/proinflammatory endothelium:** 3A. Elevated arachidonoyl-DAG *activates* **protein kinase C (PKC)** and **p38-MAPK**. *(Inferred from Lemaire; demonstrated in endothelial cells: PMID: 23542698; PMID: 25498910.)* 4A. PKC/p38-MAPK signaling *upregulates* **tissue factor and ICAM-1**, promotes endothelial activation, apoptosis, and impaired angiogenesis, *resulting in* a **prothrombotic, proinflammatory endothelial surface** — *without complement deposition*. *(Demonstrated in vitro: PMID: 25498910.)*

**Branch B — loss of endothelial protective signaling:** 3B. PIP2 shortage *impairs* **VEGFR2 signaling** (VEGFA pathway) downstream of the receptor. *(Demonstrated in vivo: PMID: 33986189.)* 4B. Failed VEGFR2→**Akt** activation *reduces* **COX-2** expression and **PGE2** production. *(Demonstrated: PMID: 33986189; PMID: 26887830.)* 5. Branches A and B converge: a prothrombotic, repair-deficient glomerular microvasculature *leads to* **microvascular/capillary thrombosis, endothelial swelling, and podocyte injury** (thrombotic microangiopathy + nephrotic-range proteinuria). 6. Glomerular TMA *results in* the **clinical**

**triad** (hemolytic anemia from mechanical RBC shearing, platelet consumption/thrombocytopenia, acute kidney injury) *plus* proteinuria, hematuria, and hypertension. 7. Recurrent injury *leads to* **chronic kidney disease** / **ESKD** over time.

## Diagram




## Supporting detail

- **Molecular pathways:** PI cycle (lipid resynthesis); DAG–PKC signaling; p38-MAPK; VEGFA–VEGFR2→PI3K–Akt; COX-2/prostaglandin (PGE2) axis.
- **Cellular processes:** endothelial activation, apoptosis, impaired angiogenesis/migration, platelet activation, podocyte dysfunction, microthrombosis.
- **Protein dysfunction:** complete loss of a constitutively active lipid kinase (catalytic/expression loss-of-function, not misfolding-aggregation).
- **Immune involvement: complement-independent** — a defining feature. Bruneau et al. showed endothelial activation and death "independently of complement activation": "loss of DGKε expression/activity in EC induces an increase in ICAM-1 and tissue factor expression through the upregulation of p38-MAPK-mediated signals" (PMID: 25498910).

- **GO term suggestions:** GO:0004143 (diacylglycerol kinase activity), GO:0006661 (phosphatidylinositol biosynthetic process), GO:0007205 (DAG/PKC signaling), GO:0038084 (VEGF signaling), GO:0007596 (blood coagulation), GO:0001525 (angiogenesis).
- **CL term suggestions:** CL:0000115 (endothelial cell), CL:1000746 (glomerular capillary endothelial cell), CL:0000653 (podocyte), CL:0000233 (platelet).

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## Section 7 — Anatomical Structures Affected

- **Primary organ:** kidney — glomerulus (UBERON:0000074), renal glomerular capillary/endothelium.
- **Secondary / systemic:** microvasculature broadly; hematologic effects (hemolysis, thrombocytopenia). Rare extrarenal TMA features may occur but renal involvement dominates.
- **Body systems:** urinary/renal; hematologic; cardiovascular (microvascular, hypertension).
- **Tissue / cell level:** glomerular endothelial cells (CL:1000746), podocytes (CL:0000653), platelets (CL:0000233). Histology shows TMA and, in a subset, MPGN-like changes (PMID: 32413569).
- **Subcellular compartments:** endoplasmic reticulum and Golgi (DGKε is S-palmitoylated at Cys38/40 and localizes to the ER; PMID: 38008259); plasma membrane (PI-cycle lipid signaling). GO cellular-component terms: GO:0005789 (ER membrane), GO:0005886 (plasma membrane).
- **UBERON suggestions:** UBERON:0002113 (kidney), UBERON:0000074 (renal glomerulus), UBERON:0004638 (renal glomerular capillary).
- **Lateralization:** bilateral (systemic genetic disease; both kidneys).

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## Section 8 — Temporal Development

- **Onset:** congenital/infantile — median ~11 months, essentially all before age 2 years (PMID: 36988693; PMID: 23542698). Onset pattern is acute (TMA episode) on a chronic genetic background.

- **Progression / course:** relapsing-remitting superimposed on a progressive decline. Relapses occurred in ~66.7% and death/CKD progression in ~60% at median 3.3-year follow-up (PMID: 36988693). A notable natural-history feature is that acute TMA episodes may diminish after early childhood while proteinuria/CKD persists.
- **Disease duration:** chronic, lifelong.
- **Remission patterns:** both spontaneous and treatment-associated remissions are reported, but relapses are common; notably, relapses can occur **as isolated nephrotic-range proteinuria** during or despite C5 blockade (PMID: 42077707; PMID: 33879077).
- **Critical period:** infancy/early childhood is the window of greatest vulnerability and the key period for diagnosis and nephroprotective intervention.

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## Section 9 — Inheritance and Population

- **Epidemiology / incidence:** ultra-rare. UK registry data defined the incidence of **DGKE aHUS at 0.009/million/year** and **DGKE "MPGN" at 0.006/million/year**, a combined **0.015/million/year** (PMID: 32386968). In early-onset (<2 years) aHUS cohorts, *DGKE* mutations account for a meaningful minority — e.g., 4/83 in the Spanish registry (PMID: 25135762).
- **Inheritance pattern:** autosomal recessive.
- **Penetrance:** high/complete for the renal phenotype in biallelic carriers (with variable expressivity across aHUS/MPGN/SRNS presentations).
- **Expressivity:** variable — same genotype can yield aHUS, MPGN-like disease, or isolated SRNS (PMID: 32413569).
- **Consanguinity:** homozygous cases in consanguineous or founder settings are reported; consanguinity increases recessive risk.
- **Anticipation / mosaicism / founder effects:** no repeat-expansion anticipation; no specific founder mutation established, though recurrent alleles (e.g., p.Trp322Ter) exist. Carrier frequency is low (ultra-rare alleles).
- **Demographics:** reported across multiple ethnicities worldwide; no strong sex bias documented for this monogenic disease. Age distribution is strongly skewed to infancy.

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## Section 10 — Diagnostics

**Clinical/laboratory signature.** The diagnostic trigger is **infantile aHUS (TMA triad) accompanied by nephrotic-range proteinuria/hematuria and typically normal complement C3**. "For infants and young children with aHUS in conjunct with nephrotic level proteinuria, variants of the DGKE gene should be screened" (PMID: 37994143).

**Core labs:** CBC with schistocytes, low haptoglobin, elevated LDH, indirect hyperbilirubinemia (hemolysis); thrombocytopenia; elevated creatinine; urinalysis (proteinuria, hematuria); serum albumin (low), cholesterol (elevated); complement C3/C4 (usually normal); ADAMTS13 activity (to exclude TTP); stool studies/Shiga toxin (to exclude STEC-HUS).

**Biopsy/pathology:** renal biopsy shows thrombotic microangiopathy; a subset shows MPGN-like glomerular changes (PMID: 32413569).

**Genetic testing (definitive).** Molecular confirmation requires ***DGKE* sequencing**, best delivered via a **targeted aHUS/TMA gene panel** that includes *DGKE* alongside complement genes (*CFH*, *CFHR1–5*, *CFI*, *CD46/MCP*, *C3*, *CFB*, *THBD*) and, where panels are negative, whole-exome sequencing (as in the original discovery). MLPA of the CFH-CFHR region is useful to characterize co-modifiers. Chromosomal microarray, karyotyping, FISH, mtDNA, and repeat-expansion testing are *not applicable*.

**Clinical criteria / differential diagnosis.** Must distinguish from (1) complement-mediated aHUS (low C3 more typical, complement variants, often eculizumab-responsive), (2) STEC/typical HUS (preceding bloody diarrhea, Shiga toxin), (3) TTP (ADAMTS13 <10%), and (4) secondary TMA. The combination of infantile onset + heavy proteinuria + normal C3 points to *DGKE* (PMID: 37994143).

**Screening.** Cascade genetic testing of at-risk siblings and carrier testing of parents (recessive); no population newborn screen exists.

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## Section 11 — Outcome / Prognosis

- **Renal outcomes:** guarded. ~60% death or progression to CKD and ~67% relapse at median 3.3 years (PMID: 36988693); persistent hypertension/proteinuria in ~40%.

- **Natural history nuance:** acute TMA episodes may attenuate after early childhood, but proteinuria and progressive CKD persist, so long-term prognosis is dominated by nephropathy rather than recurrent hemolytic crises (PMID: 32386968).
- **Mortality:** disease-specific mortality is low relative to morbidity; the dominant adverse outcome is ESKD requiring dialysis/transplant.
- **Transplant:** because the defect is intrinsic to kidney cells (endothelium/podocytes) and is **not** circulating-complement-mediated, post-transplant TMA recurrence risk is expected to be lower than in complement-mediated aHUS — an important prognostic/therapeutic distinction (mechanistic inference supported by the cell-intrinsic, complement-independent biology; PMID: 25498910).
- **Prognostic factors:** degree/persistence of proteinuria, hypertension control, baseline GFR, relapse frequency, and co-inherited complement variants (worse with complement modifiers; PMID: 25135762).

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## Section 12 — Treatment

**Overarching principle:** DGKE aHUS is **complement-independent**, so complement blockade is often ineffective at preventing relapses; genotype-guided care emphasizes nephroprotection and de-escalation of unnecessary anti-C5 therapy.

- **Complement inhibition (eculizumab / ravulizumab; NCIT: Eculizumab C88248).** Widely used empirically at presentation before genotype is known. Evidence is mixed: some patients with low C3 appear to respond (PMID: 33879077; PMID: 25135762), but relapses occur **despite** C5 blockade in the absence of complement activation — "eculizumab did not prevent disease relapses, which occurred in the absence of complement activation" (PMID: 42077707). Genetic confirmation supports discontinuation/de-escalation to avoid overtreatment: "The lack of genetic testing options has led to overtreatment in this patient" (PMID: 42077707).
- **Nephroprotection (first-line for the chronic phase).** RAAS blockade (ACE inhibitors/ARBs) for proteinuria and hypertension; blood-pressure control; supportive care for CKD.
- **Plasma therapy.** Plasma infusion/exchange has controlled recurrences in some patients, particularly those with co-existing complement/THBD variants (PMID: 25135762).

- **Antithrombotic strategies.** Given the prothrombotic mechanism (tissue factor/PKC), antiplatelet/anticoagulant approaches are a rational consideration, though not formally validated.
- **Mechanism-based / experimental.** Restoration of the PGE2 axis is the leading targeted concept: a **stable PGE2 analog reversed TMA and proteinuria** in endothelial-specific *Dgke*-knockout mice (PMID: 33986189). This is preclinical; no human trial yet.
- **Renal replacement / transplant.** Dialysis and kidney transplantation for ESKD; recurrence risk appears lower than complement-mediated aHUS (see Prognosis).
- **Pharmacogenomics:** none established.

**NCIT term suggestions:** Eculizumab (C88248), ACE Inhibitor (C705)/Angiotensin Receptor Antagonist, Plasmapheresis/Plasma Exchange (C50785), Kidney Transplantation (C15366), Dialysis (C15248).

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## Section 13 — Prevention

- **Primary prevention:** not possible for a monogenic recessive disease beyond reproductive counseling. **Genetic counseling** for consanguineous/carrier families; options include carrier testing, prenatal testing, and preimplantation genetic diagnosis (recessive, 25% recurrence risk per pregnancy).
- **Secondary prevention:** early recognition of infantile aHUS with heavy proteinuria and prompt *DGKE* sequencing to enable early nephroprotection and to avoid overtreatment with complement inhibitors (PMID: 37994143; PMID: 42077707); cascade testing of siblings.
- **Tertiary prevention:** aggressive BP/proteinuria control to slow CKD; relapse surveillance; **meningococcal vaccination** is required for any patient placed on eculizumab (anti-C5 therapy increases meningococcal risk; PMID: 27811836).
- **Public-health / environmental interventions:** not applicable.

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## Section 14 — Other Species / Natural Disease

- **Orthologous gene:** mouse *Dgke* (NCBI Gene 56077); orthologs conserved across vertebrates. DGKε's arachidonoyl specificity and PI-cycle role are evolutionarily conserved (PMID: 27803897).
- **Naturally occurring disease in other species:** no naturally occurring DGKE nephropathy has been documented in companion animals or wildlife (OMIA search not productive). *Not available.*
- **Comparative biology:** engineered mouse models (below) faithfully reproduce the endothelial TMA mechanism, supporting conservation of the pathway.
- **Zoonotic potential / transmission:** none (genetic disease).

## Section 15 — Model Organisms

| Model | Type | Key phenotype | Recapitulation | Reference |
|---|---|---|---|---|
| Endothelial-specific *Dgke*-knockout mouse | Mammalian, conditional KO | TMA + proteinuria; impaired VEGFR2→Akt→COX-2→PGE2; **reversed by stable PGE2 analog** | High — reproduces renal TMA and identifies druggable axis | PMID: 33986189 |
| Global *Dgke*-knockout mouse | Mammalian, constitutive KO | Endothelial distress; impaired glomerular Cox-2 and PGE2 production | Moderate–high (mechanistic) | PMID: 26887830 |
| DGKε-knockdown human endothelial cells | In vitro cellular | ↑ICAM-1/tissue factor via p38-MAPK; ↑apoptosis; impaired migration/angiogenesis; **no complement deposition** | Mechanism of complement-independent EC injury | PMID: 25498910 |
| Recombinant purified DGKε / DGKεΔ40 | In vitro biochemical | Confirms arachidonoyl specificity, constitutive activity | Enzyme-function model | PMID: 28199087 |

**Model applications:** dissecting the endothelial VEGFR2–PGE2 axis, testing PGE2-based and antithrombotic therapeutics, and validating complement-independence. **Limitations:** mouse models may not fully capture the human proteinuric/podocyte spectrum or the aHUS/MPGN/SRNS phenotypic overlap; in vitro knockdown lacks organ-level hemodynamics. **Resources:** MGI (*Dgke*), standard conditional-KO repositories.

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## Key Findings (with evidence)

### Finding 1 — Recessive, complement-independent Mendelian disease caused by biallelic *DGKE* loss-of-function

Exome sequencing identified recessive *DGKE* mutations co-segregating with aHUS in nine unrelated kindreds, and the authors stated they "identified recessive mutations in DGKE (encoding diacylglycerol kinase ε) that co-segregated with aHUS in nine unrelated kindreds, defining a distinctive Mendelian disease" (PMID: 23542698). DGKε has no role in complement; variants span missense, frameshift, nonsense, and splice classes; a 12-patient cohort reported 13 biallelic variants (7 pathogenic, 3 likely pathogenic, 11 novel) (PMID: 36988693). Lemaire explicitly inferred that "loss of DGKE function results in a prothrombotic state."

### Finding 2 — Mechanism: prothrombotic endothelium via DAG/PKC + collapse of VEGFR2→COX-2→PGE2 signaling

DGKε is arachidonoyl-DAG specific and recycles phosphatidylinositol; its loss elevates arachidonoyl-DAG, activating PKC/p38-MAPK and upregulating tissue factor and ICAM-1 (PMID: 23542698; PMID: 25498910). In endothelial-specific *Dgke*-KO mice, "loss of Dgke in the endothelium results in impaired signaling downstream of VEGFR2 due to cellular shortage of phosphatidylinositol 4,5-biphosphate," and "treating the endothelial specific Dgke-knockout mice with a stable PGE2 analog was sufficient to reverse the clinical manifestations of thrombotic microangiopathy and proteinuria" (PMID: 33986189).

## Finding 3 — Infantile-onset nephropathy with proteinuria and progression to CKD

"Patients presented at a median age of 11 (7.5, 12.3) months; all were younger than 2 years," and at median 3.3-year follow-up "significant hypertension and/or proteinuria (40%), relapses (66.7%), and death or progression to CKD (60%) were common" (PMID: 36988693). Lemaire: affected individuals "present with aHUS before age 1 year, have persistent hypertension, hematuria and proteinuria (sometimes in the nephrotic range), and develop chronic kidney disease with age" (PMID: 23542698).

## Finding 4 — Ultra-rare; eculizumab efficacy controversial

"We defined the incidence of DGKE aHUS as 0.009/million/year and so-called DGKE MPGN as 0.006/million/year, giving a combined incidence of 0.015/million/year" (PMID: 32386968). "Eculizumab did not prevent disease relapses, which occurred in the absence of complement activation" (PMID: 42077707).

## Finding 5 — Models recapitulate endothelial TMA and are pharmacologically rescuable

Endothelial-specific KO mice develop TMA/proteinuria reversible by a PGE2 analog (PMID: 33986189); global KO mice show impaired glomerular Cox-2/PGE2 (PMID: 26887830); DGKε-knockdown endothelial cells activate without complement deposition (PMID: 25498910).

## Finding 6 — DGKε is a uniquely arachidonoyl-specific, constitutively active PI-cycle lipid kinase

"It is the only DGK isoform with specificity for a particular species of DAG, i.e., 1-stearoyl-2-arachidonoyl glycerol" (PMID: 27803897); "DGKε is involved in the PI cycle maintaining the cellular level of phosphorylated PI derivatives of signaling activity" (PMID: 38008259). Its substrate selectivity (LOX-like motif) explains the disease's lipid-signaling selectivity (PMID: 22266092; PMID: 28199087).

### Finding 7 — Diagnostic signature + genotype-guided management

Infantile onset + TMA + nephrotic-range proteinuria/hematuria + normal C3 + MPGN-like histology → sequence *DGKE*: "For infants and young children with aHUS in conjunct with nephrotic level proteinuria, variants of the DGKE gene should be screened" (PMID: 37994143). Genetic diagnosis prevents overtreatment with complement blockade — "the lack of genetic testing options has led to overtreatment in this patient" (PMID: 42077707).

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## Mechanistic Model / Interpretation

The disease is best understood as a **cell-intrinsic lipid-signaling failure** of the glomerular microvasculature. DGKε sits at a single, non-redundant node in the phosphatidylinositol cycle, uniquely handling the arachidonoyl-DAG that is the PI backbone. When that node fails, two consequences propagate in parallel: (i) a **gain of a prothrombotic/proinflammatory program** (DAG→PKC/p38-MAPK→tissue factor/ICAM-1, with endothelial apoptosis), and (ii) a **loss of a protective program** (PIP2 depletion→VEGFR2→Akt→COX-2→PGE2). The convergence of "more thrombosis, less repair" on the glomerular endothelium and podocyte explains the simultaneous TMA *and* heavy proteinuria that make this entity clinically distinctive. The **absence of complement involvement** is the organizing clinical fact: it explains the typically normal C3, the frequent failure of eculizumab to prevent relapses, and the rationale for genotype-guided de-escalation. The model is strongly supported because the critical protective node (PGE2) is both necessary (lost in KO mice) and sufficient (its pharmacologic restoration reverses disease) — an unusually complete causal demonstration for a rare disease.

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## Evidence Base

| PMID | Role | How it supports / challenges findings |
|---|---|---|
| 23542698 | Landmark human-genetics [H] | Defines DGKE as recessive causal gene; infers prothrombotic LOF mechanism |
| 36988693 | Nationwide cohort [H] | Quantifies onset (median 11 mo), phenotype frequencies, outcomes |
| 32386968 | UK registry [H] | Incidence 0.015/million/yr; long-term outcomes, treatment response |
| 25135762 | Spanish registry [H] | Complement modifiers influence onset/severity; plasma/complement therapy useful with co-variants |
| 33986189 | Endothelial KO mouse [M] | Establishes VEGFR2→PGE2 mechanism; PGE2 analog reverses disease |
| 26887830 | Global KO mouse [M] | Impaired glomerular Cox-2/PGE2 corroborates mechanism |
| 25498910 | In vitro endothelial [V] | Complement-independent EC activation via p38-MAPK/tissue factor/ICAM-1 |
| 27803897 | Biochemistry review [V/C] | Unique arachidonoyl substrate specificity |
| 38008259 | Biochemistry [V] | S-palmitoylation; ER localization; PI-cycle role |
| 28199087 | Protein purification [V] | Confirms acyl specificity; links LOF to aHUS |
| 22266092 | Mutagenesis [V] | LOX-like motif governs acyl-chain selectivity |
| 42077707 | Case report [H] | Eculizumab fails to prevent relapses; supports de-escalation |
| 33879077 | Case report [H] | Low-C3 patient apparently eculizumab-responsive — nuance/challenge |
| 32413569 | Case series [H] | aHUS/SRNS phenotypic overlap; variable histology |
| 37994143 | Case/guidance [H] | Diagnostic trigger: infantile aHUS + nephrotic proteinuria → sequence DGKE |
| 38374836 | Case report [H] | DGKE + CFI heterozygous variants — modifier concept |

## Limitations and Knowledge Gaps

- **Small sample sizes.** All clinical data derive from small case series/registries (largest cohorts ~12 patients); frequencies and outcome percentages carry wide uncertainty.
- **Phenotype–genotype correlation is incomplete.** Why the same *DGKE* genotype produces aHUS vs MPGN vs isolated SRNS is unresolved; modifier contributions (complement variants, triggers) are only partly characterized.
- **Eculizumab evidence is anecdotal and conflicting.** No randomized data exist; apparent responses in low-C3 patients may reflect co-modifiers.
- **PGE2 therapy is preclinical.** The compelling mouse rescue has not been translated to humans; dosing, delivery, and safety in infants are unknown.
- **Transplant recurrence risk** is inferred from mechanism rather than established by large transplant cohorts.
- **QoL, natural history beyond childhood, and adult outcomes** are poorly documented.

## Proposed Follow-up Experiments / Actions

1. **International DGKE nephropathy registry** with harmonized genotype, histology, complement status, treatment, and long-term renal outcomes to firm up frequencies and genotype–phenotype correlations.
2. **Prospective / structured evaluation of complement-inhibitor withdrawal** in genetically confirmed DGKE patients, to formalize de-escalation guidance.
3. **Translational development of PGE2-axis therapeutics** (stable PGE2 analogs or EP-receptor agonists) toward first-in-human studies, building on the KO-mouse rescue.
4. **Mechanistic dissection of the aHUS/MPGN/SRNS branch point**, including podocyte-specific *Dgke* models and single-cell profiling of glomerular cell types.
5. **Evaluate antithrombotic/PKC-directed strategies** in model systems given the tissue-factor/PKC arm of the mechanism.
6. **Define post-transplant recurrence risk** through multicenter transplant-cohort analysis.

*Evidence source types: human clinical (PMIDs 23542698, 36988693, 32386968, 25135762, 32413569, 42077707, 33879077, 37994143, 38374836); model organism (33986189, 26887830); in vitro cellular (25498910); in vitro biochemical/structural (27803897, 38008259, 28199087, 22266092).*

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